Energy xxx (2011) 1–6

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Forecasting the differences between various commercial oil prices in the Persian Gulf region by neural network

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ARTICLE INFO

Article history:
Received 25 January 2011
Received in revised form
26 April 2011
Accepted 4 May 2011
Available online xxx

Keywords:
Persian Gulf
Oil price
Neural network
Forecasting

ABSTRACT

In this paper we have investigated the differences between the prices of different commercial oils of the Persian Gulf region. The prices of 7 different crude oils from Iran, Kuwait, Saudi Arabia, Oman, Abu Dhabi and Dubai were compared with the benchmark light oil of Saudi Arabia over the period January 2000 –April 2010. A neural network is introduced to forecast the price of any commercial oil in these crude oils, provided that the price of the benchmark light oil of Saudi Arabia is already known or is predicted by another forecasting method. The designed neural network is able to predict the differences in the oil prices with an average error of 8.82% for testing and 7.24% for training data. It is claimed that the present method can promote the forecasting power of existing models to predict the price of any commercial oil instead of an average or benchmark value.

1. Introduction

Crude oil is still world's main source of energy, providing about 66% of the global energy demand and consisting more than 10% of world's trade [1]. Over 160 different crude oils are internationally traded. All of these crude oils have different qualities and prices. The price of crude oil is of great importance to the stability and growth prospects of world economy [2,3]. For example it is declared that a 10% change in the oil price may cause a 2.5–6% fluctuation in the gross domestic product of the United States [4]. Continuous and sometimes serious surges and falls in the oil prices affect the global economy and every body's life. Numerous studies have been conducted on this matter and the amount of researches and public concern usually rise after any shock in the crude oil price [5]. Factors that affect the price of a certain kind of oil may be classified into three main groups.

The first group consists of factors that depend on the crude oil quality. Crude oils have various physical and chemical properties such as density, chemical composition, sulfur, viscosity, heating value, boiling range, asphaltene content and so on. These properties affect different processes such as oil transportation, refining and various petrochemical processes. It is customary in the oil market to qualify the crude oils according to their density and sulfur content. Lighter crude oils are usually priced higher due to easier refining and higher yields of valuable products such as gasoline. Sulfur content has a strong negative effect on the crude oil price because an additional cost of sulfur removal is necessary for crudes of high sulfur content. These factors are not responsible for the surges and rises in the oil prices. They are responsible for the price differences between different types of crude oils like the Brent crude of North Sea or the West Texas Intermediate crude of the United States. Although the quality of oil produced from a certain oil field changes with time, these changes are always very steady and may be easily ignored when other much stronger time dependent factors are considered. As most researchers (and also most ordinary people) are more interested in changes in the oil prices as a whole (and not the differences between the different crude oils), these factors are usually ignored in common oil price foretelling and analysis.

Economic and market oriented factors are generally considered to be the main causes of surges and falls in the global oil prices. Many researchers have tried to describe and even predict the changes in the oil prices in terms of different economic factors. All of these researches have considered the oil price as a whole and ignored the differences between the different commercial oils [6,7].

Third group of factors consist of geopolitical considerations which may cause very sharp changes in the oil prices. Political tensions and even wars in oil rich areas such as the Middle East, Nigeria or Venezuela may cause sharp rises or even shocks in the oil market [8]. It is generally accepted that the most important long run factor that controls the oil price is the relationship between global supply and demand. While demand for oil depends on the

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doi:10.1016/j.energy.2011.05.004

economic conditions of the developed and developing countries (group 2 factors), the supply of oil is strongly affected by the geopolitical considerations (group 3 factors). Both of these groups of factors are highly time dependent, but of course the economic factors vary slowly and gradually, while political factors usually have a sharp but temporary effect. Changes of the economic factors with time are also more predictable and may be achieved by the analysis of the economic data by professionals, but political analyzers are usually less accurate in forecasting of the political affairs. Numerous researchers have tried to forecast the oil prices, partly because this issue is also very interesting for the oil market and the investors.

Barone-Adesi et al. [9] used semi empirical equations to forecast the oil prices. Noel [10] investigated the impact of the oil price on the production of crude oil and gas while Yasnder [11] studied the relation between oil price and economic stability. Gulen [12] used the continuity principle to predict the price of the West Texas Intermediate (WTI) crude oil. The GARCH model was used by Morana [13] to forecast the short-term oil prices. Mirmirani and Li [14] forecasted the oil prices in the American market by means of different mathematical methods including neural networks and genetic algorithms. Lanza et al. [15] used error correction models to predict the oil price. Abramson and Finizza [16] tried to predict the oil price by means of a neural network model. Xie et al. [17] used support vector machine method to forecast the oil price. Rehrl and Friedrich [18] modeled long-term oil price with a Hubbert approach. Gori et al. [19] forecasted the oil price and consumption under three different scenarios.

The models introduced in the literature are very different and usually consider an average value of oil or are based on the prices of some important benchmark crude. Modeling and finding the relations between prices of different commercial crude oils have been ignored by all of these researchers. In this paper we aim to construct a model to predict the differences between the prices of the different crudes, provided that the price of the benchmark crude is available. Differences between the prices of different crudes are more reasonable and usually depend on some more or less permanent factors such as crude quality and geological location, so oil price differences may be more predictable than the average oil price itself. This research group also believes that this kind of modeling may fill an existing gap in this area and lead to more realistic and reliable oil price forecasting.

2. Theory

To construct a general framework for predicting the price of a certain kind of crude oil, we may consider the oil price as an addition of two different terms. The first term may be described as the global average oil price which is unique for all internationally traded crudes and the other would be a difference term that varies for each kind of crude oil. The above concept may be shown mathematically as below:

$$(Oilpr)_i = \overline{Avpr} + (Devpr)_i \tag{1}$$

Where $(Oilpr)_i$ is the price of a certain kind of oil in a certain time, $\overline{Avpr}$ is the global average price of the oil in a certain time and $(Devpr)_i$ is the difference term which differs for each kind of oil. In this research, we aim to model the difference term or $(Devpr)_i$ of the different crudes. Both of these terms depend on time, but of course the first term's dependence is much stronger. When the oil market feels severe changes, the differences between the prices of different oils may also vary, but the variation would not be as sharp as the changes in the average price of oil itself. It seems that $\overline{Avpr}$ is more related to permanent economical and political factors while $(Devpr)_i$ which is unique for each kind of oil is more dependent on the oil quality. In fact this quantity also depends on the geographical, economic and political factors and also may change with time. Political tensions may cause serious changes in the crude oil price of one or several countries. At the other hand the geological location of the crude oil sources may also affect the oil prices. It is clear that the oil market classifies crude oils according to their production areas such as the Persian Gulf, the North Sea and so on. It seemed too idealized to predict whole of the global oil prices by such a simple framework, so we decided to restrict the research to a single oil producing area and try to model the differences in the oil prices according to a benchmark crude in that area. The equation (1) was then modified as below:

$$(Oilpr)_i = BMOilpr_A + (DevprA)_i \tag{2}$$

Where $(Oilpr)_i$ is the price of a certain kind of oil in a certain time in that area, $BMOilpr_A$ is the price of the benchmark oil in the certain time in that area and $(DevprA)_i$ is the difference term of the price of the certain crude oil that makes it different from the price of the benchmark crude. Limiting the research to a certain oil producing area my also has other benefits. Although even the political, economic and geological situations may vary considerably among the different countries of the same area, the differences are not as sharp as the differences between these countries and other parts of world. For example consider the Persian Gulf area. Of course the economic, political and cultural situations of Iran, Saudi Arabia and Iraq are different, but their differences are much less than their differences with the United States, China and the European Union countries. At the other hand the geological differences of the countries in a certain area and the factors that affect their access to oil markets may be more similar to each other. So it was concluded that the term $(DevprA)_i$ in the equation (2) may be more reasonable and predictable than the term $(Devpr)_i$ in equation (1). To limit the numerical range of data, both sides of the equation (2) were also divided by the term $BMOilpr_A$ to reach the following equation.

$$(ROilprA)_i = 1 + (RDevprA)_i \tag{3}$$

Where $(ROilprA)_i$ is the ratio of the price of a certain kind of oil to the price of the benchmark oil on a given time and $(RDevprA)_i$ is the ratio of the difference term to the price of benchmark oil. As the variations of these new terms are much smaller and less time dependent, the mathematical analysis may be more accurate. We have tried to predict the value of $(RDevprA)_i$ as a function of time, source and quality of oil.

In this study we have selected the Persian Gulf oil producing countries as our target area and light oil of Saudi Arabia as the benchmark crude. Average monthly prices of the Persian Gulf crude oils consisting of light and heavy crudes of Iran, light and heavy crudes of Saudi Arabia, crude oils of Oman, Dubai, Abu Dhabi and Kuwait were collected from January of 2000 until April of 2010. API density and the sulfur percentage have been considered as the measures of oil quality. These properties are universally accepted in global oil market. The deviations of the different oil prices in the Persian Gulf area were computed according to the benchmark light oil of Saudi Arabia.

3. Artificial neural network modeling

ANN (Artificial neural network) models are mathematical models, akin to vast network of neurons in the human brain [20,21]. Artificial neural network (ANN) is a mathematical tool, which tries to represent low-level intelligence in natural organisms and it is a flexible structure, capable of making a non-linear mapping between input and output spaces [22]. Neural networks have

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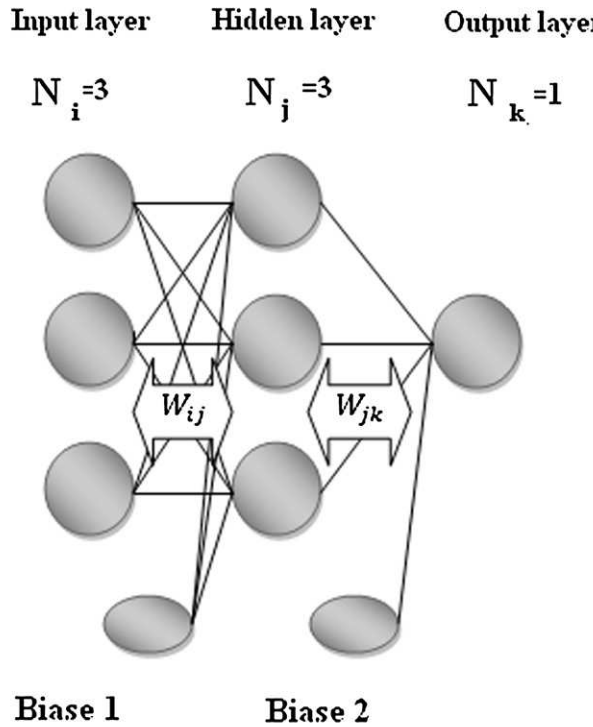

Fig. 1. General structure of a neural network.

received considerable attention in various fields of research such as statistical science, engineering, computer science, artificial intelligence, energy consumption [23–26], solar radiation [27,28], rate of heat transfer [29] and many others [30–32]. The artificial neural network (ANN) is a well-known tool for solving complex, non-linear biological systems [33,34] and it can give reasonable solutions even in extreme cases or in the event of technological faults [35]. There are many different ANN architectures like MLP (multi-layer perceptron), RBF (radial basis function) and RNN (recurrent neural network). Each of these architectures has been used for modeling of different case studies. In this study we have used the multi-layer feed forward neural network which is described in the next paragraph.

These ANNs, like others, consist of multiple layers (input layer, hidden layer or layers and output layer). Information flows from

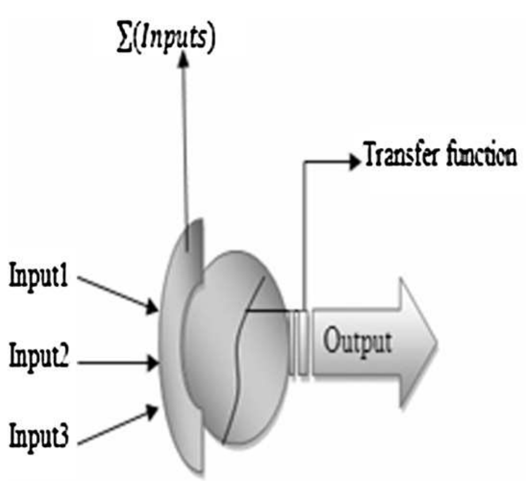

Fig. 2. General structure of a neuron.

Table 1
Ranges of parameters and details of scaling parameters [eia.doe.gov, 2010; API, 2010; Mitchellorg, 2010].

Actual value	Range	*m*	*c*
API	27–39	0.0667	−1.7000
Sulfur content	0.8–2.8	0.4000	−0.2200
Country code	1–5	0.2000	−0.1000
Time	0.0833–10.3333	0.0780	0.0935
Oil price ratio	0.7997–1.3102	1.5670	−1.1531

input to output layer through hidden layers. Fig. 1 shows a typical multi-layer feed forward ANN which consists of three neurons in each of the input, hidden and output layers. Except from the neurons in the input layer, all other neurons receive a collection of inputs and produce a single output with aid of transfer function. The overall structure of a typical neuron is shown in Fig. 2. A feed-forward neural network was designed using back-propagation training algorithm. The number of neurons in input and output layers depends on the independent and dependent variables, respectively. Since one dependent variable (oil price) depends on four dependent variables of (API density, sulfur content, country and time), four and one neurons were devoted to input and output layers, respectively. The number of hidden layers and their neurons depend on the complexity of the problem to be investigated. In this study a hidden layer including 15 neurons was used and the activation function was selected to be logarithmic sigmoid. The mean relative percent error (P) between the predicted and desired values was calculated to show the performance of the network.

4. Input data

Input data used in this work consisted of two parts. First part includes information about the oil properties which are taken from the American Petrochemical Institute [36,37] and the second part consists of oil prices which are taken from the American Energy management Agency [38]. These oil prices consist of FOB prices of different crudes on the exporting ports of the Persian Gulf in a 124 months period (From January 2000 until April 2010). These data are gathered for five main exporting oil countries of the Persian Gulf (Iran, Saudi Arabia, Kuwait, Oman and the Emirates). The crude oils which were studied in this work included light and heavy crudes of Iran, light and heavy crudes of Saudi Arabia and crude oils of Oman, Dubai, Abu Dhabi and Kuwait (8 main exporting crude oils of the Persian Gulf area). A total of 3968 data were obtained and

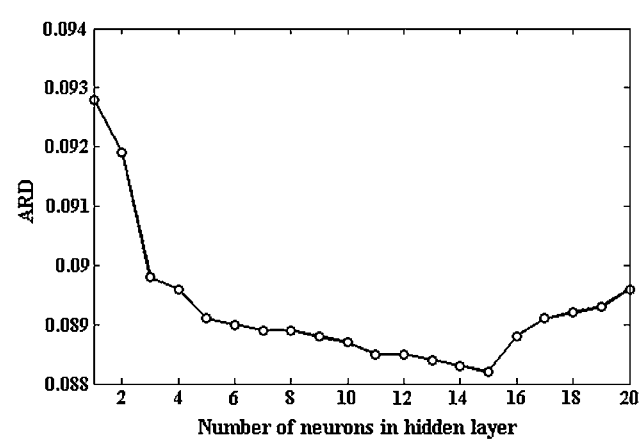

Fig. 3. Deviation of absolute relative deviation (ARD) with number of neurons in hidden layer.

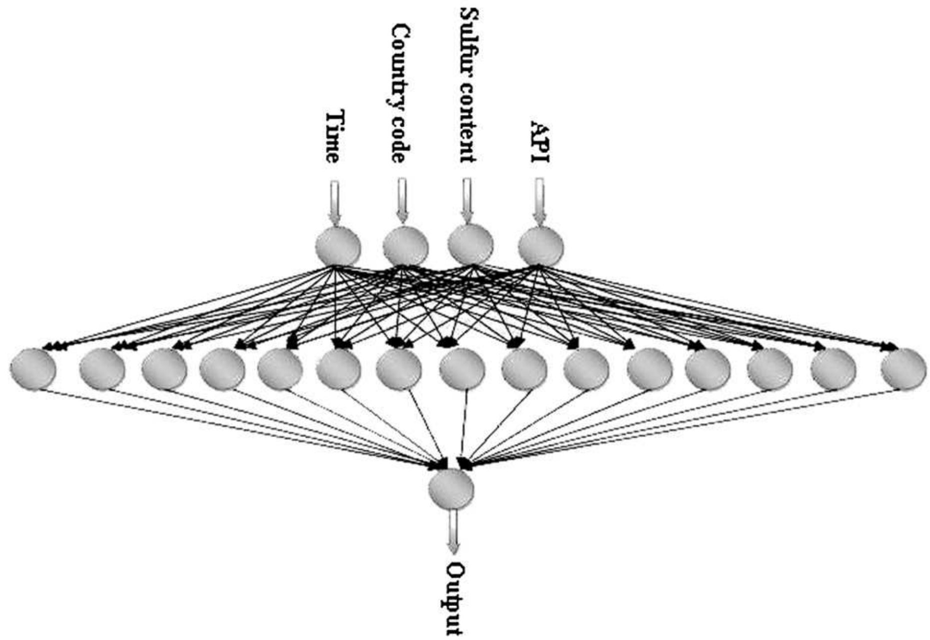

Fig. 4. General structure of the neural network used in this work.

approximately 60% of them were used for training of the ANN, 10% for validation and remaining 30% for testing of the neural network. The original data were scaled according to the following formula to increase the convergence speed and minimize the errors [39].

$$(\text{Scaled value}) = (\text{Actual value}) \times m + c \tag{4}$$

In this work the scaling range is limited to [0.1–0.9] range for all of the input and output parameters. The details of the scaling parameters are shown in the Table 1. 60% of the data were used to train and 10% were used to validate the ANN and then the network was tested with the remaining 30% of data. A part of the training data was used for validation. In order to keep track of the validation error, random samples from the training data set were used to form the validation set of each application, while the size of the validation set was equal to 10% of the total data set. The validation error decreases at the beginning of the training, but at some point it starts to increase even though the training error still decreases. The best parameters of the network were those of a minimum validation error, so the learning was stopped at that point to avoid the over-fitting of the data.

The average percent error which is obtained by equation (5) was also calculated as a measure of agreement between model predictions and actual values.

$$P = \frac{1}{N} \times \sum_{i=1}^{N} \left| \frac{\left(RoilprA_i^{act} - RoilprA_i^{pred} \right)}{RoilprA_i^{act}} \right| \tag{5}$$

Where $RoilprA_i^{act}$ and $RoilprA_i^{pred}$ stand for the predicted and actual oil price ratios, N stands for the number of data and P is the average percent error. The number of neurons in the hidden layer is known to have a sharp effect on the model precision, so the number of neurons where changed to find the optimum value. The result is illustrated in Fig. 3 which clearly shows that the best answer would be obtained by considering around 15 neurons in the hidden layer. The graphical presentation of the final ANN model is shown in the Fig. 4. This figure shows that we have used a three layer ANN with

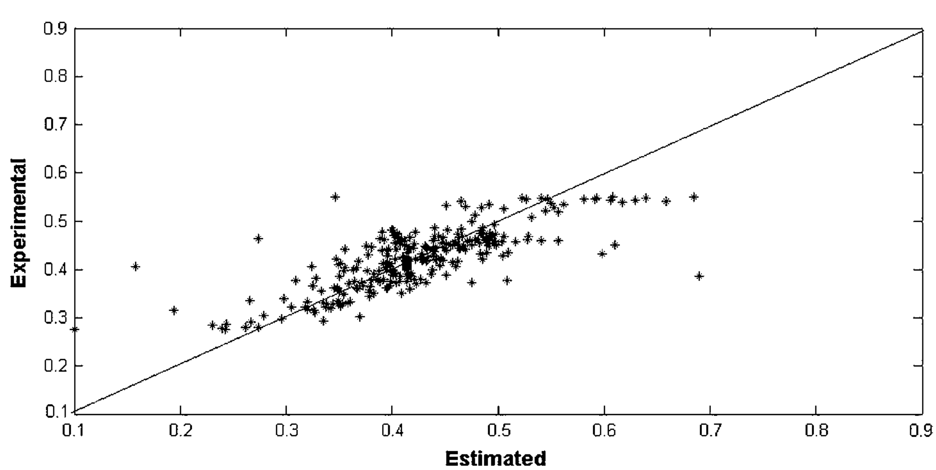

Fig. 5. Comparison of actual and forecasted values for testing data.

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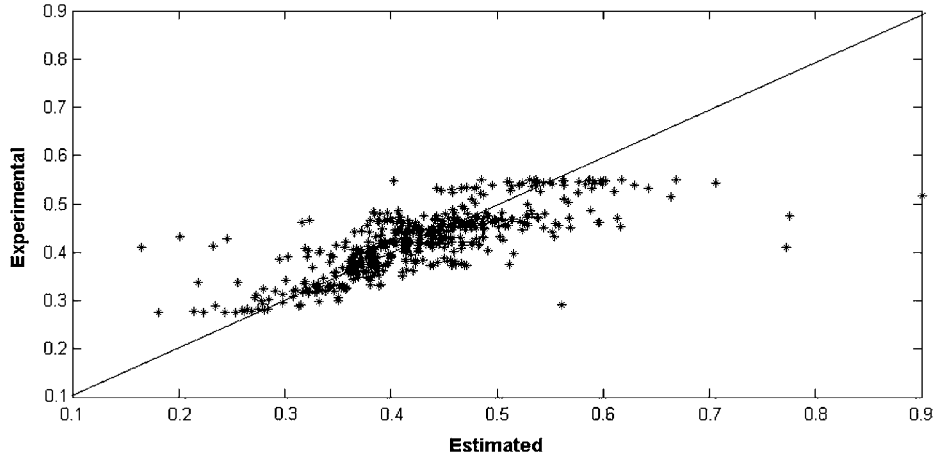

Fig. 6. Comparison of actual and forecasted values for training data.

four inputs neurons (API gravity, sulfur content, country code and time), a single output neuron (oil price ratio) and 15 neurons in the hidden layer.

5. Results and discussions

After training and evaluation of ANN, its ability to predict the oil price ratios has to be checked. In this work the predictions of the proposed architecture are compared with both of the testing and training data in Figs. 5 and 6, respectively. The diagonal lines in the middle of these figures are the locations of exact predictions and points in the figure show training or testing data. The error of each prediction is proportional to the distance between each point and the diagonal line, so the cluster of points near the diagonal line may illustrate the overall accuracy of the present model. The mean percent error for testing and training data is also calculated to be 8.82% and 7.24%, respectively. These values refer to the differences between actual and forecasted differences between the prices of different commercial oils and the price of the benchmark crude. Values of ARD (average relative deviation), MSE (mean square error), MAD (mean absolute deviation) and MAE (mean absolute error) for different crudes of the Persian Gulf region are reported in Table 2.

The present model may be criticized for some good reasons. In fact the model is not able to predict the price of any kind of commercial oils. Given the value of the benchmark oil in a given time, the model has the ability to produce good estimations of other commercial oils in that area. Although the estimation of oil price differences is not considered as important as the average oil price, it seems to be more predictable.

The present model is not considered to be an alternative or substitution to any of the existing models. In fact this model can serve as a companion to any of the previous models to extend their abilities to forecast the price of different commercial oils in a certain oil producing area. The present model would be able to predict the price of some important commercial oils of the Persian Gulf region, provided that the price of the benchmark oil (light oil of Saudi Arabia) is predicted by another method. Of course we think that the present model may improve the ability and generality of other existing methods and fill an existing gap for obtaining a general and reliable model to forecast the price of any commercial oil. Since the input variables of the ANN are also selected to be simple and available for any commercial oil, coupling of the present model with any other methods for prediction of the average or benchmark prices is not so difficult. The forecasted benchmark or average prices together with the characteristics of the commercial oil are simply given to the ANN and the hybrid model is able to estimate the different prices for different commercial oils.

Table 2
Values of ARD, MSE, MAD and MAE for different crudes in the Persian Gulf region.

	ARD (%)	MSE	MAD	MAE
Light crude of Iran	6.320	0.0016	0.3131	0.0316
Heavy crude of Iran	6.272	0.0017	0.0249	0.0251
Light crude of Saudi Arabia	1.113	0.00003	0.0037	0.0046
Heavy crude of Saudi Arabia	13.520	0.0026	0.0359	0.0351
Dubai	10.840	0.0043	0.0448	0.0447
Kuwait	8.735	0.0018	0.0300	0.0302
Abu Dhabi	9.898	0.0052	0.0493	0.0498
Oman	9.761	0.0048	0.0443	0.0443

6. Conclusions

The method presented in this study is able to promote the forecasting power of the existing oil price prediction models. This approach consists of the design of an appropriate neural network to predict the differences between the different oil prices in a specified oil producing region, provided that the benchmark oil price is known or predicted by another forecasting method. So the existing forecasting models would be able to predict the price of any commercial oil instead of the average or benchmark value. The vicinity of the present model is limited to the commercial oils of the Persian Gulf region but it is possible to propose similar ANN architectures for other oil producing areas.

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